

# **Development Automation & Linux Lab Test**

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# LAB TEST

1. Write a program to print the patterns. (Use loop)

a. 1

2 3

4 5 6

7 8 9 10

```
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ cat onea.sh
#!/bin/bash

flag=1
for (( i=1; i<=4 ; i++));
do
    for (( j=1; j<=i; j++));
    do
        echo -n $flag
        ((flag=flag+1))
    done
    echo ""
done
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ ./onea.sh
1
23
456
78910
```

b. 7 8 9 10

4 5 6

2 3

1

```
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ cat oneb.sh
#!/bin/bash

flag=10
for(( i=3; flag>0; i-- ));
do
    ((temp=flag-i))
    for(( j=0; j<=i; j++ ));
    do
        echo -n $temp
        ((temp=temp+1))
        ((flag=flag-1))
    done
    echo ""
done
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ ./oneb.sh
78910
456
23
1
```

c. 1  
2 3  
4 5 6  
7 8  
9

```
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ cat onec.sh
#!/bin/bash

flag=1
for(( i=0; i<3; i++ ));
do
    for(( j=2-i; j>0; j-- ));
    do
        echo -n " "
    done
    for(( k=0; k<=i; k++));
    do
        echo -n $flag" "
        ((flag=flag+1))
    done
    echo ""
done

for(( i=0; i<2; i++));
do
    for(( j=0; j<=i; j++ ));
    do
        echo -n " "
    done
    for(( k=0; k<2-i; k++));
    do
        echo -n $flag" "
        ((flag=flag+1))
    done
    echo ""
done
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ ./onec.sh
1
2 3
4 5 6
7 8
9
```

d. 1,1,2,3,5,....

```
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ cat oned.sh
#!/bin/bash

a=0
b=1
c=0
echo -n $b
for(( i=0; i<8; i++));
do
    ((c=a+b))
    echo -n ,$c
    ((a=b))
    ((b=c))
done
echo ""
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ ./oned.sh
1,1,2,3,5,8,13,21,34
```

2. Create the pattern of the above questions and move it to another folder

```
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ ./onea.sh; ./oneb.
sh; ./onec.sh; ./oned.sh; > output
1
23
456
78910
78910
456
23
1
  1
  2 3
4 5 6
  7 8
  9
1,1,2,3,5,8,13,21,34
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ mv output ./dir1
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ ls dir1
output
```

3. Write a make file of question1 and move it to another folder

```
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ cat makefile
myexecutable: onea.sh oneb.sh onec.sh oned.sh
    cat $^ > myexecutable
    chmod +x myexecutable
    mv myexecutable ./dir2

clean:
    rm -f myexecutable
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ make
cat onea.sh oneb.sh onec.sh oned.sh > myexecutable
chmod +x myexecutable
mv myexecutable ./dir2
```

```
ayush@ayroid-Victus:/mnt/Storage/College/SEM4/Linux-Lab/test$ ./dir2/myexecutabl
e
1
23
456
78910
78910
456
23
1
  1
  2 3
4 5 6
  7 8
  9
1,1,2,3,5,8,13,21,34
```